Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 2

Full Marks: 20 *Deadline: 11:59 PM, 06-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Execute all the commands in Linux and take the screenshots.
3) Prepare a PDF that contains the question number and corresponding screenshots of the outputs.
4) The naming of the file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, your roll number is B24CS055, then the filename should be:
   B24CS055_Shareen_02.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Getting Familiar with Text Editors.
2) Getting real-time experience with text processing commands.
3) Leaning file and text manipulation with the help of text processing.
4) Understanding the regular expressions and their need in the manipulation of texts.

Task 1: Overview of Sort, Grep, Awk and Sed [10 Marks]

[Instruction: If the file is being modified, you have to show the contents of the file with the cat command]

Create a directory named students_marks. Enter inside the directory and create the file named marksheet.txt having [roll_no, name, subject, marks]. The fields of each record should be separated by ", " (Comma spaced). Use the following data to create the file:

Roll No	Name	Math	Physics	Chemistry
101	Alice	85	78	91
102	Bob	72	65	70
103	Charlie	88	82	90
104	David	66	74	69
105	Eve	93	89	95
106	Frank	58	60	62
107	Grace	81	84	79
108	Hannah	75	68	73
109	Ivan	69	71	74
110	Jack	90	86	88

File should be created like:
101, Alice, Math, 85
102, Bob, Math, 72
…
110, Jack, Math, 90
101, Alice, Physics, 78
…
110, Jack, Physics, 86
101, Jack, Chemistry …

1. Create the file and show the 10th line to the 20th line of the file. [1 Marks]

2. Change all the ‘, ’ to ‘::’ of the file, and marksheet.txt should be modified with it. [1 Marks]
3. Calculate everyone’s total marks from marksheet.txt (You are not allowed to calculate the sum manually; use the awk command to calculate the sum) and append it to another file named total_marks.txt. Format should be like this: (roll_no, name, total_marks_obetained, total_marks)
   101, Alice, 254, 300
   …
   110, Jack, 264, 300 [2 Marks]
4. From total_marks.txt, print roll_no, name and grade with the given scenario:
   a. 90% and above: Grade A
   b. 80% and above: Grade B
   c. 70% and above: Grade C
   d. 60% and above: Grade D
   e. 50% and above: Grade E
   Your output should look like the format: roll_no::name::grade
   Example: 101::Alice::grade [2 Marks]
5. Delete all the records of **Frank** from both marksheet.txt and total_marks.txt. Files should be updated with the modified records. [2 Marks]
6. Sort total_marks.txt with respect to the total marks obtained field (third field) in decreasing order of marks and modify the file with the sorted result. [1 Marks]
7. Change Eve’s name to Eva in both marksheet.txt and total_marks.txt files, and modify the file with the changed result. [1 Marks]

Task 2: Difference, Compare, Cut and Grep, Sort [10 Marks]

[Instruction: If the file is being modified, you have to show the contents of the file with the cat command]

Create a directory named dir2. Inside that directory construct a file with name student_id.txt where put the following data with ‘::’ having format id::rollno:

Id	Name
1002	Shukla
1003	Devi

1005	Nath

Format:
1002::Sukla
1003::Devi
1005::Nath
Now create another file with name student_age.txt where put the following data with "::" having format id::age:

ID	**Age**
1005	23
1002	21
1003	18

Format:
1005::23
1002::21
1003::18

1. Now join these two tables based on ids append the result to a new file id_name_age.txt [2 Marks]
2. Now sort the id_name_age.txt with respect to age (Third field) in increasing order. [1 Marks]
3. Show only ID and Age of all records. [1 Marks]
4. Create another file, student_id_new.txt and copy the contents of the student_id.txt file to this new file. Now change the ID of Sukla to 1023 using any method you prefer. Now compare these two files side by side. [2 Marks]
5. Come out of the directory. Find all the files with the "23" string, not a substring. [1 Marks]
6. Again, sort the student_id_new.txt with the dictionary of the order of names (second field) [2 Marks]
7. Compare student_id_new.txt and student_id.txt. If the files are the same, print "Same"; otherwise, print "Different". [1 Marks]